

# Supply Chain Risk Report — Carbon Fiber Price Surge Contingency

**Date:** 2026-05-20  
**Report ID:** SCR-2026-0520-001  
**Product:** Pokee 碳素超轻钓竿 (SKU-1)  
**Trigger:** Toray T1100 Carbon Fiber +30% price increase

## 1. Executive Summary

Toray Japan has announced a **30% price increase** on T1100 carbon fiber, the primary material for the main shaft of our flagship product **Pokee 碳素超轻钓竿 (SKU-1)**. This report quantifies the cost impact and outlines alternative supplier options to mitigate supply chain risk.

## 2. Current BOM — SKU-1 (Pokee 碳素超轻钓竿)

| Component  | Material                 | Weight (g) | Unit Cost (USD) | Lead Time (Days) | Supplier          |
|------------|--------------------------|------------|-----------------|------------------|-------------------|
| Main Shaft | Toray T1100 Carbon Fiber | 120        | \$85.00         | 45               | Toray Japan       |
| Guides     | Titanium Alloy + SIC     | 30         | \$45.00         | 30               | Fuji Guides       |
| Handle     | AAA Grade Cork           | 25         | \$15.00         | 15               | Portugal Cork Co. |
| Total      |                          | 175        | \$145.00        |                  |                   |

## 3. Cost Impact Simulation (+30% Toray T1100)

| Component          | Original Cost | New Cost | Delta             |
|--------------------|---------------|----------|-------------------|
| Main Shaft (T1100) | \$85.00       | \$110.50 | +\$25.50          |
| Guides             | \$45.00       | \$45.00  | \$0.00            |
| Handle             | \$15.00       | \$15.00  | \$0.00            |
| Total              | \$145.00      | \$170.50 | +\$25.50 (+17.6%) |

**Key Finding:** A 30% material cost increase on the main shaft drives a **17.6% total product cost increase** (\$25.50 per unit).

## 4. Alternative Supplier Recommendations

### 4.1 Carbon Fiber Substitutes for T1100

| Alternative                                      | Grade               | Est. Unit Cost | Cost vs. T1100 | Lead Time | Risk Level | Notes                                                                     |
|--------------------------------------------------|---------------------|----------------|----------------|-----------|------------|---------------------------------------------------------------------------|
| <b>Teijin T800</b><br>(Teijin Co., Japan)        | High modulus        | ~\$78          | -8.2%          | 40 days   | Low        | Mature supplier, slightly lower tensile strength (5,200 MPa vs 6,370 MPa) |
| <b>SGL T800S</b> (SGL Carbon, Germany)           | High performance    | ~\$82          | -3.5%          | 35 days   | Low        | EU-based, shorter lead time, IATF 16949 certified                         |
| <b>HexTow AS4S</b><br>(Hexcel, USA)              | Aerospace grade     | ~\$90          | +5.9%          | 30 days   | Medium     | Premium pricing but stable long-term contracts available                  |
| <b>Guangzhou Dayu CF</b> (Dayu Composite, China) | Domestic high-grade | ~\$55          | -35.3%         | 20 days   | High       | Significant cost savings but requires full QA re-validation               |
| <b>Zhongwen Carbon</b><br>(Zhongwen Co., China)  | T700-class          | ~\$50          | -41.2%         | 15 days   | High       | Lowest cost option, suitable for budget-tier product line                 |

### 4.2 Recommended Strategy

| Timeline                       | Action                                                                                 | Owner       |
|--------------------------------|----------------------------------------------------------------------------------------|-------------|
| <b>Immediate (0-2 weeks)</b>   | Negotiate volume-based pricing with Toray Japan; lock current rates for 6-month supply | Procurement |
| <b>Short-term (2-4 weeks)</b>  | Request samples from Teijin T800 and SGL T800S for performance testing                 | R&D / QA    |
| <b>Medium-term (4-8 weeks)</b> | Initiate qualification process for Dayu Composite as cost-reduction backup             | Procurement |

| TIMELINE               | ACTION                                                                               | OWNER        |
|------------------------|--------------------------------------------------------------------------------------|--------------|
| Long-term (8-12 weeks) | Establish dual-source supply chain: Toray (premium) + Chinese supplier (budget tier) | Supply Chain |

## 5. Financial Impact Summary

| SCENARIO                           | UNIT COST | MARGIN IMPACT     | ANNUAL IMPACT (EST. 10,000 UNITS) |
|------------------------------------|-----------|-------------------|-----------------------------------|
| Baseline (current)                 | \$145.00  | —                 | —                                 |
| +30% T1100 only                    | \$170.50  | -25.5 pts         | -\$255,000                        |
| Switch to Teijin T800              | \$148.50  | -3.5 pts          | -\$35,000                         |
| Switch to Dayu CF                  | \$119.50  | -25.5 pts savings | +\$255,000                        |
| Dual-source (60% Toray / 40% Dayu) | \$140.20  | +4.8 pts          | +\$48,000                         |

## 6. Risk Matrix

| RISK                                 | LIKELIHOOD | IMPACT   | MITIGATION                          |
|--------------------------------------|------------|----------|-------------------------------------|
| Toray extends price hike beyond 30%  | Medium     | Critical | Dual-source strategy                |
| Alternative supplier quality failure | Medium     | High     | Staged qualification, pilot batches |
| Lead time disruption (geopolitical)  | Low        | High     | Safety stock 3-month buffer         |
| Contract renegotiation delays        | High       | Medium   | Parallel-track supplier onboarding  |

## 7. Approval & Next Steps

- [ ] Procurement to initiate Toray renegotiation by **2026-05-27**
- [ ] R&D to receive Teijin/SGL samples by **2026-06-10**
- [ ] QA to complete alternative material testing by **2026-06-30**
- [ ] Executive sign-off on dual-source strategy by **2026-07-15**